CHAPTER 11

# Spectroscopy

## Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

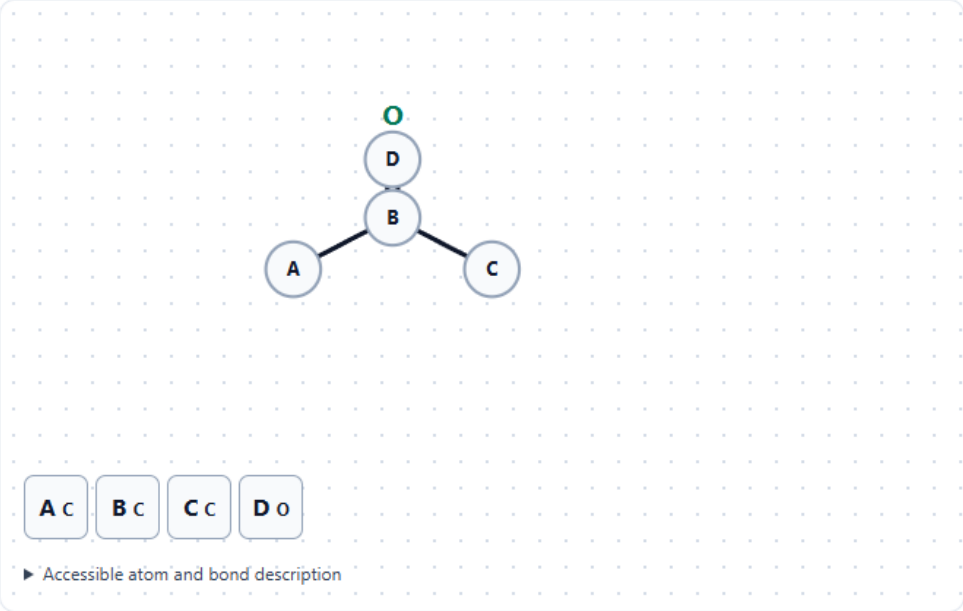
Source snapshot SHA-256: 62bb9823874ba4729b5d48335a400ea6e22a2b685ec0cc1e2fb8bd9893db1f38

BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

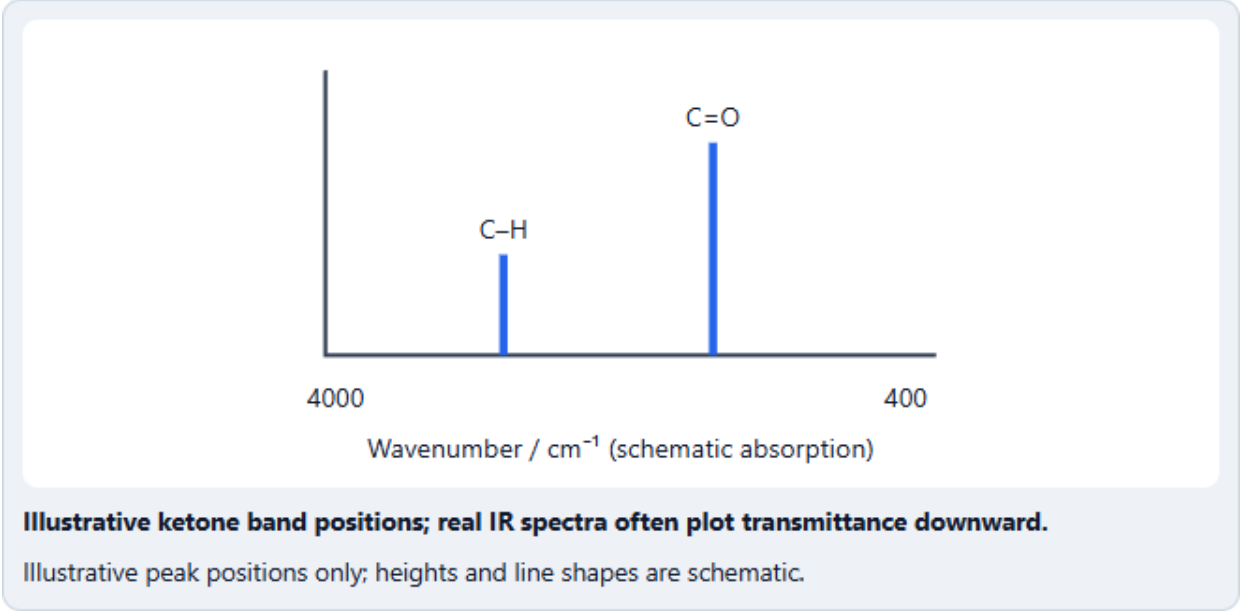
Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

# ir-read | Independent exercise

Screenshot: ir-read-stage-1-molecule




Screenshot: ir-read-diagram




{"diagramSpecification": {"type": "spectrum", "min": 400, "max": 4000, "unit": "Wavenumber / cm$^{-1}$ (schematic absorption)", "caption": "Illustrative ketone band positions; real IR spectra often plot transmittance downward.", "peaks": [{"x": 1715, "height": 0.95, "label": "C=O"}, {"x": 2950, " height": 0.45, "label": "C–H"}]}}

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

## ir-read:stage-1 | Select the bond endpoints responsible for the prominent carbonyl band.

Answer type: atoms

## ir-read:stage-2 | An IR-active vibration must change…

Answer type: choice

Option 0 (A): Dipole moment

Option 1 (B): Molecular formula

Option 2 (C): The number of carbon atoms

## ir-read:stage-3 | A broad 3200–3600 $cm^{-1}$ band appearing after reduction supports…

Answer type: choice

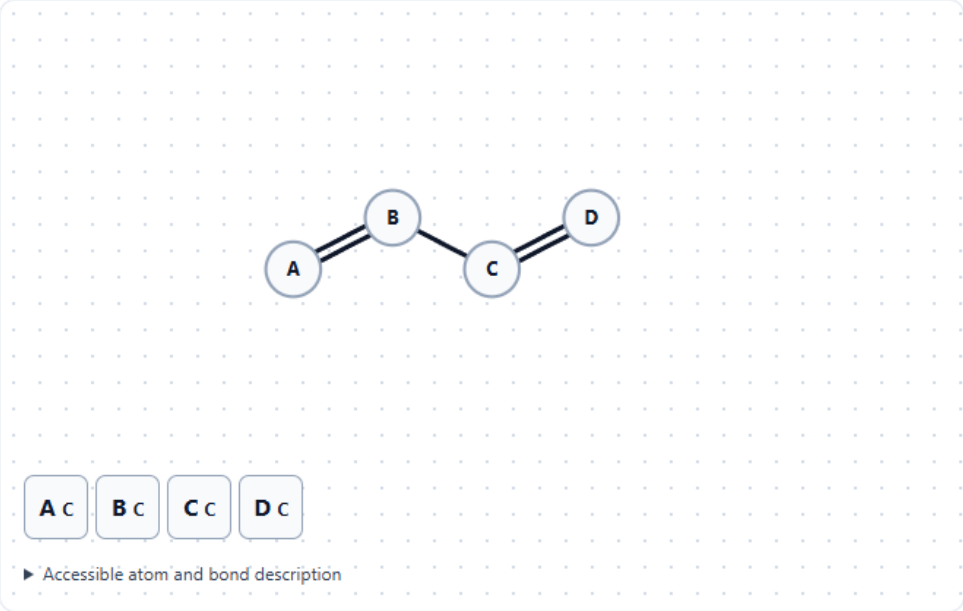
Option 0 (A): An alcohol OH

Option 1 (B): Only a new alkane

Option 2 (C): A nitrile as the sole explanation

# uv-energy | Independent exercise

Screenshot: uv-energy-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 2]]

## uv-energy:stage-1 | Select all carbons participating in the conjugated diene.

Answer type: atoms

## uv-energy:stage-2 | For comparable pi systems, increased conjugation often shifts absorption to…

Answer type: choice

Option 0 (A): Longer wavelength

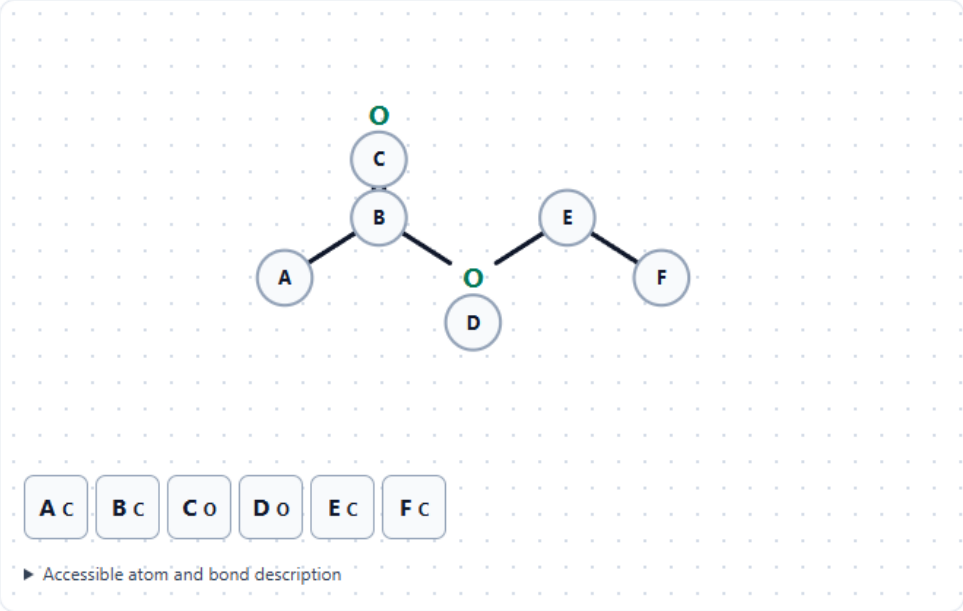
Option 1 (B): Shorter wavelength in every case

## uv-energy:stage-3 | If ε=10000 L $mol^{-1}$ $cm^{-1}$, l=1 cm, and c=0.000020 mol $L^{-1}$, what is A?

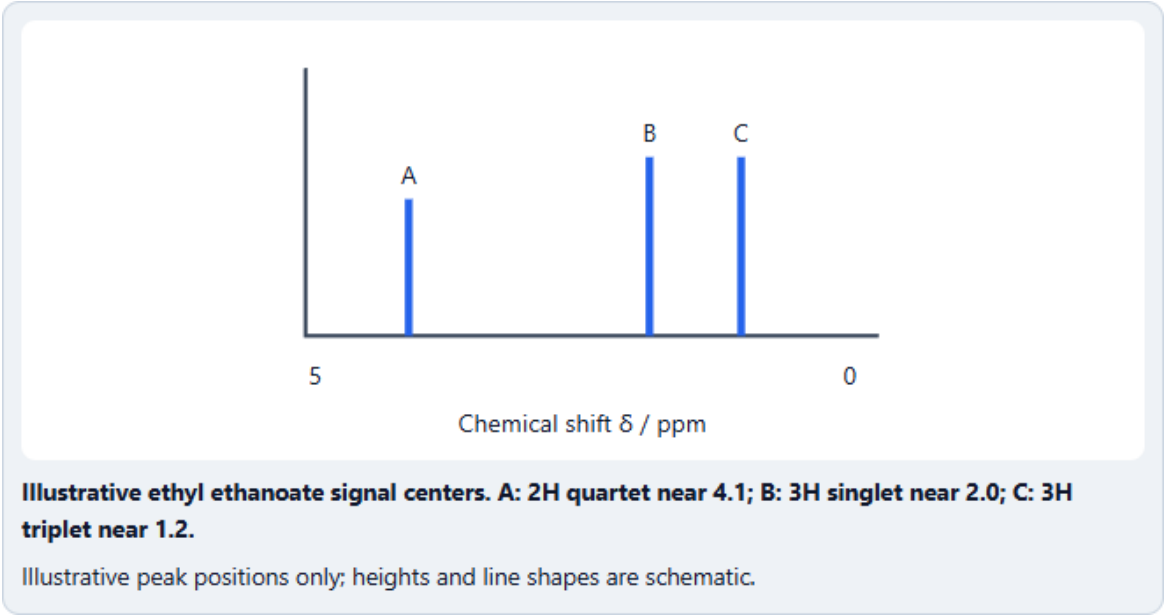
Answer type: number | unit: dimensionless

# nmr-ethyl | Independent exercise

Screenshot: nmr-ethyl-stage-1-molecule




Screenshot: nmr-ethyl-diagram




**Illustrative ethyl ethanoate signal centers. A: 2H quartet near 4.1; B: 3H singlet near 2.0; C: 3H triplet near 1.2.**

Illustrative peak positions only; heights and line shapes are schematic.

{"diagramSpecification": {"type": "spectrum", "min": 0, "max": 5, "unit": "Chemical shift δ / ppm", "caption": "Illustrative ethyl ethanoate signal centers. A: 2H quartet near 4.1; B: 3H singlet near 2.0; C: 3H triplet near 1.2.", "peaks": [{"x": 4.1, "height": 0.65, "label": "A"}, {"x": 2, "height": 0.85, "label": "B"}, {"x": 1.2, "height": 0.85, "label": "C"}]}}

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=45.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=100.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=100.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=155.00, y=120.00; charge=0; explicit H override=none

4 (E): C; x=210.00, y=85.00; charge=0; explicit H override=none

5 (F): C; x=265.00, y=120.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1], [3, 4, 1], [4, 5, 1]]

## nmr-ethyl:stage-1 | Select the OCH2 carbon.

Answer type: atoms

## nmr-ethyl:stage-2 | How many lines does its signal have in the simple n+1 model?

Answer type: number | unit: lines

## nmr-ethyl:stage-3 | The terminal ethyl CH3 gives an approximate…

Answer type: choice

Option 0 (A): Triplet, integrating to 3H

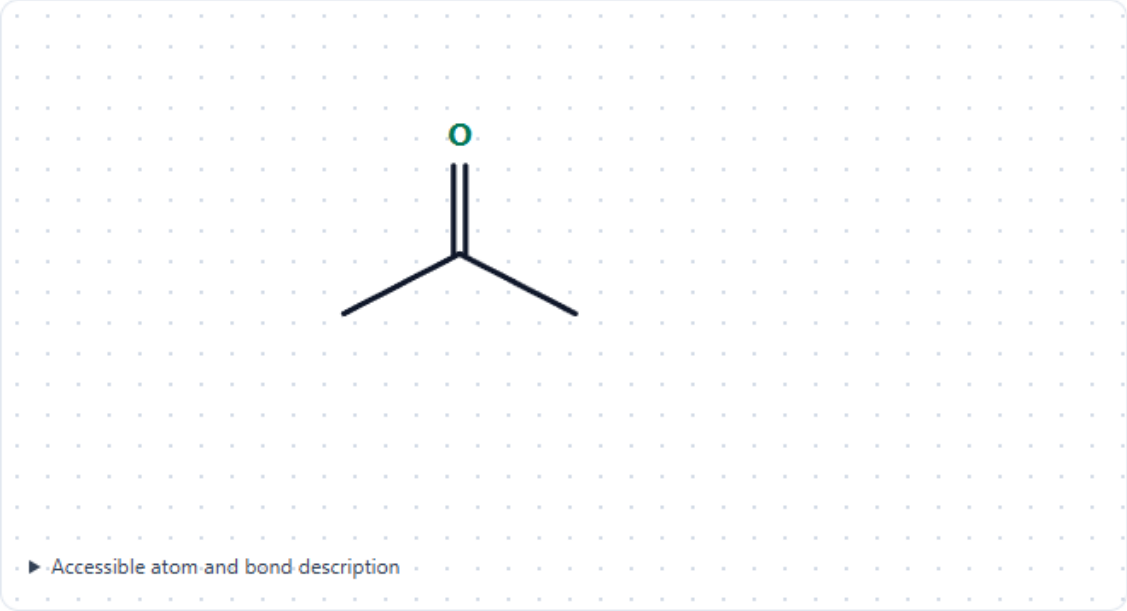
Option 1 (B): Quartet, integrating to 2H

Option 2 (C): Singlet, integrating to 1H

# combined-data | Independent exercise

Supplied illustrative data: C3H6O; a ketone-like C=O band; one proton NMR singlet integrating to 6H near 2 ppm.

Screenshot: combined-data-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

## combined-data:stage-1 | For formula C3H6O, calculate DBE.

Answer type: number | unit: DBE

## combined-data:stage-2 | A single 6H singlet near 2 ppm suggests...

Answer type: choice

Option 0 (A): Two equivalent methyl groups with no adjacent H-bearing carbon

Option 1 (B): Six nonequivalent one-H environments

Option 2 (C): An aldehydic proton at 2 ppm

## combined-data:stage-3 | With a ketone-like IR band, name the consistent C3H6O structure.

Answer type: text